

Arijit De

Kolkata, India

+918981800329 | arijitde2050@gmail.com | <https://github.com/arijitde92> | <https://www.linkedin.com/in/de-arijit/>

PROFESSIONAL SUMMARY

Machine Learning Engineer with a PhD having 6 years of industry experience spanning Multi-Agent AI systems, Computer Vision and MLOps. Designed and built scalable, robust multi modal RAG pipelines, 3D/2D Image Segmentation and Object Detection solutions across construction, healthcare, and automotive domains. Proficient in Python, FastAPI, Langchain, CrewAI, PyTorch and cloud deployment (GCP, AWS).

WORK EXPERIENCE

Machine Learning Engineer | mVizn, Singapore | Remote JAN 2024 – PRESENT

- Led development of a multi-agent platform for construction domain (**CrewAI, Python, FastAPI**) powering Autonomous Project Management & Scheduling (APMS), and intelligent project information retrieval — serving 100 concurrent users / 3000 agent runs per day and increasing business efficiency by 50%.
- Hardened multi-agent workflows for reliability by adding failure-mode handling (retries, timeouts, tool-call fallbacks, and output **guardrails**), reducing task failure rate by 60%.
- Instrumented end-to-end observability, tracing, and performance monitoring with **CrewAI Tracing** and **Ragas**, lifting evaluation scores (faithfulness / answer-relevancy) by 75%.
- Built multi-modal Retrieval-Augmented Generation (**RAG**) pipelines over massive text, image, and CAD datasets (10M+ vectors in **Pinecone**) using hybrid search and cross encoder reranking, achieving retrieval accuracy (recall@k) of 92% while holding retrieval latency under 600 ms.
- Applied **LoRA fine-tuning** to adapt Gemini LLMs for domain understanding and structured output, raising structured-output validity to 98% and cutting token/inference cost per request by 30%.
- Used Google Cloud Platform (**GCP**) services like Cloud Storage, Cloud Run, IAM, RBAC to deploy and maintain dynamic scalability with 99% uptime.

Machine Learning Engineer | Synerjix, USA | Remote JUN 2025 – DEC 2025

- Built a Smart Moving Assistant that estimates shifting cost of items from room photos, improving stakeholder business efficiency by 50%, adopted by 3500 users / 10000 estimates processed.
- Deployed DinoV3-based object detection at 65.3 mAP, serving inference at 10ms via **Python + FastAPI, MySQL, and AWS** (SageMaker, EC2, EKS) with CI/CD and **Docker**.

Computer Vision Engineer | Institute of Neurosciences, India | Onsite JAN 2022 – DEC 2023

- Designed and implemented custom 3D Computer Vision solutions for Neurological disorder analysis (**PyTorch, Python, OpenCV, Numpy**), improving diagnosis speed by 40% and reaching 88.68% accuracy on clinical scans.
- Implemented **MLflow**-based experiment tracking, model versioning, model registry, and reproducible training to support scalable **MLOps**, cutting model iteration/turnaround time by 50%.
- Optimized models with **ONNX** and deployed via Docker and **GCP** (Cloud Storage, Compute Engine VMs, IAM, Cloud SQL), reducing inference latency by 65%.

Machine Learning Engineer | Mercedes-Benz Research & Development India, India | Onsite AUG 2018 – SEP 2019

- Developed and optimized a computer vision pipeline for Vulnerable Road User (VRU) detection by fine-tuning **YOLOv3** on large-scale annotated datasets, attaining detection mAP50 of 83% at real-time inference speeds.
- Designed & deployed model training, evaluation, and validation workflows in **AWS**.
- Built automated quality assurance (**QA**) and evaluation frameworks using Python, improving annotation throughput by 20% and data quality by 25%.

Systems Engineer | Tata Consultancy Services, India | Onsite AUG 2014 – APR 2015

- Maintained high-availability backend systems (Java, MySQL) for TCS's internal website, ensuring seamless functionality for over 100,000 users. Streamlined feature release cycle, reducing deployment time by 20%.

PROJECTS

Doctor Appointment Scheduler | https://github.com/arijitde92/Doctor_Appointment_Scheduler

- Built a multi-agent conversational appointment-booking system (Google ADK) with a root coordinator delegating to doctor-matching, review, and scheduling agents.
- Integrated **MCP**-based tool access for structured data and external actions — Cloud SQL MCP for doctor/clinic/availability lookups, Google Maps MCP for travel-time estimation, and Google Calendar MCP (via Composio) for appointment booking.
- Implemented grounded retrieval over doctor review data using Vertex AI RAG Engine and Gemini 2.5 Flash for conversational reasoning.
- Deployed end-to-end on Google Cloud Run.

EDUCATION

PhD Jadavpur University, India	FEB 2020 – APR 2025
M.Tech. Computer Science & Engineering Jadavpur University, India GPA - 8.83	AUG 2016 – JUN 2018
B.Tech. Computer Science & Engineering Techno India, India GPA - 8.81	AUG 2010 - MAY 2014

SKILLS

Languages:	Python, C++, SQL
LLM/Agent Systems:	Multi-Agent Orchestration (CrewAI, Google ADK, LangGraph), Agent Evaluation (Ragas, DeepEval), LLM Guardrails, Observability/Tracing (CrewAI Tracing), MCP, Agentic Memory, LangChain, LLM APIs (OpenAI, Gemini, Claude)
Frameworks / Libraries:	FastAPI, Flask, PyTorch, Scikit-learn, OpenCV, Open3D, PyTest
Cloud & Ops:	Google Cloud Platform (GCP), Amazon Web Services (AWS SageMaker, EC2, EKS), Git, Docker, MLflow, GitHub Actions